

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The function $f(t) = 40,000(2)^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

Answer

- A. 2
- B. 790
- C. 1,580
- D. 40,000

Correct Answer: B

Rationale

Choice B is correct. It’s given that t minutes after an initial observation, the number of bacteria in a population is $40,000(2)^{\frac{t}{790}}$. This expression consists of the initial number of bacteria, **40,000**, multiplied by the expression $2^{\frac{t}{790}}$. The time, in minutes, it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{790}}$ to double. Since the base is **2**, the expression $2^{\frac{t}{790}}$ will double when the exponent increases by **1**. Since the exponent of this expression is $\frac{t}{790}$, the exponent will increase by **1** when t increases by **790**. Therefore, the time, in minutes, it takes for the number of bacteria in the population to double is **790**.

Choice A is incorrect. This is the base of the exponent, not the time it takes for the number of bacteria in the population to double.

Choice C is incorrect. This is the number of minutes it takes for the population to double twice.

Choice D is incorrect. This is the number of bacteria that are initially observed, not the time it takes for the number of bacteria in the population to double.